

Michael Arbel | Research Scientist in Machine Learning

Inria Grenoble – Rhône-Alpes – Grenoble – France

✉ michael.n.arbel@gmail.com • [github michaelarbel.github.io](https://github.com/michaelarbel) • [in michael-arbel-0a38a655](https://inria.fr/people/michael-arbel-0a38a655)
[MichaelArbel](https://www.linkedin.com/in/michaelarbel)

Research profile

I develop principled machine learning methods that combine data with structure, simulators, and implicit constraints for reliable scientific inference and decision-making. My research connects statistical learning theory, optimization, and empirical work, with current directions in functional bilevel optimization, predictive representation learning, and reliable tabular learning.

Academic positions

Research Scientist (Chargé de recherche), Inria
THOTH team

Grenoble, France
Since 2022

Starting Research Fellow at Inria
Working with Julien Mairal on bilevel optimization

Grenoble, France
2021-2022

Education

PhD in Machine Learning, University College London
Under the supervision of Arthur Gretton, Pass with no revisions
Dissertation: Regularization and Optimization of Generative Adversarial Networks.

London, UK
2016-2021

M.S. in applied mathematics at École Normale Supérieure
Machine learning and computer vision (MVA), High honors
Master's thesis at École Normale Supérieure de Paris under the supervision of Stéphane Mallat,
Subject: Wavelet analysis of long-range dependencies in simple grammars.

Cachan, France
2014-2015

B.S. & M.S. in applied mathematics at École Polytechnique
Minor in physics, GPA: 3.93/4
Master thesis at Princeton University, under the supervision of René Carmona,
Subject: Mean field games with a dependence on the distribution of the control.

Palaiseau, France
2011-2014

Industrial positions

Research Engineer at Prophesee
Computer Vision for neuromorphic cameras

Paris, France
2015-2016

Grants and scholarships

2027-2032: DELPHI, ERC Starting Grant, 1.5M euros. Awarded in 2026; scheduled to begin in October 2027. Implicitly constrained learning for scientific discovery and decision-making.

2024-2028: BONSAI, ANR JCJC, 374K euros. Project coordinator. Bilevel Optimization for Simulation-based Inference; March 2024–February 2028.

2021-2022: Starting research grant from INRIA, **10K euros.**

2016-2021: Scholarship from Gatsby Computational Neuroscience Unit for the PhD program. 22K pounds/year

2011-2014: Scholarship from the Eiffel Excellence Scholarship program. 14K euros/year.

Honors and Awards

2024: Spotlight presentation at NeurIPS 2024, *Functional Bilevel Optimization for Machine Learning*.

2022: Long oral presentation at AISTATS 2022.

2021: Long oral presentation at ICML 2021.

2020: Spotlight presentation at ICLR 2020.

2018: Best Poster Award at MSR AI Summer School, Cambridge.

2014: Award of the Financial Risk Chair of Polytechnique for the Master's thesis.

Services to the Community

Editorial Activities

ICLR 2023 and NeurIPS 2023

Area chair

2023

Journal of Machine Learning Research (JMLR)

Board of reviewers

2020–present

NeurIPS (2018–2021), ICLR (2019–2021), ICML (2021).

Reviewer

2018–2021

Organization of scientific events

UCL

London, UK

DeepMind/CSML Seminars

2017–2019

Weekly research seminars in Machine Learning with invited speakers from the UK and Europe. Annual budget: 5K pounds.

Research supervision

Current PhDs: Supervision or co-supervision of Kenta Vert (2024–ongoing), Eloïse Touron (2024–ongoing), Fares El Khoury (2024–ongoing), and Pierre-Louis Ruhlmann (2023–ongoing).

PhD alumni: Ieva Petrulionyte (2023–2026), co-supervised with Julien Mairal; Juliette Marrie (2022–2025), co-supervised with Julien Mairal and Diane Larlus.

Master's: Tristan Martin (2026); Eloïse Touron and Camille Touron (2024); Ieva Petrulionyte (2023).

Research topics, collaborators, and representative student publications: michaelarbel.github.io/team/.

Teaching graduate courses

Université Grenoble Alpes

Grenoble, France

Co-instructor, *From Basic Machine Learning Models to Advanced Kernel Learning*

2026–2027

Joint IASD, MVA, and MASH course

Dauphine–PSL / ENS Paris-Saclay

Co-instructor, *Machine Learning with Kernel Methods*

Spring 2026

Earlier teaching: Kernel methods at ENS Paris-Saclay and advanced kernel learning at Université Grenoble Alpes since 2022; Intelligent Systems at Université Grenoble Alpes (2023–2024).

Selected research software

MLXP: Main author. Python framework for reproducible machine learning experiments: launching, logging, and querying results. ACM REP 2024.

EquiTabPFN: Target-permutation equivariant prior-fitted network for tabular learning. NeurIPS 2025.

funcBO: Functional bilevel optimization algorithms and applications. NeurIPS 2024 (Spotlight).

BGS-opt: Research code for AmIGO and Bilevel Games with Selection, including hyperparameter optimization and dataset distillation. ICLR 2022; NeurIPS 2022.

Selected Invited Talks

Upcoming invitations.....

19 Oct. 2026: *When Predictive Self-Supervised Learning Becomes Canonical Correlation Analysis*. Gatsby Computational Neuroscience Unit, London.

13 Oct. 2026: *What Structure Does Predictive Self-Supervised Learning Recover?* Institute for Mathematics of Toulouse.

Recent presentations.....

29 June 2026: *Bilevel optimization: A functional perspective with application to self-supervised learning*. SciCADE, Edinburgh; minisymposium on bilevel optimization for scientific computing.

17 June 2026: *PEIRA: Learning Predictive Encoders through Inter-View Regressor Alignment*. Optimization and Learning Day, LJK, Grenoble.

1 June 2026: *Bilevel optimization: A functional perspective with application to self-supervised learning*. SIAM Conference on Optimization.

21 Apr. 2026: *Learning Predictive Embeddings through Inter-View Regressor Alignment*. CILVR seminar, Center for Data Science, New York University.

Mar. 2026: *A functional approach to differentiable programming in machine learning*. Workshop on Flows on Measure Spaces and Applications in Machine Learning, Oberwolfach.

Feb. 2025: *Functional bilevel optimization: theory and algorithms*. RKHS Seminars, METU.

Invited talks at conferences and workshops.....

2023: Bayes Comp. Levi, Finland.

2023: Conference in Mathematics and Image Analysis. Berlin, Germany.

2022: Learning and Optimization in Luminy workshop. Luminy, France.

2022: ELISE Theory Workshop on Machine Learning Fundamentals. EURECOM, Sophia Antipolis, France.

2020: Workshop on Functional Inference and Machine Intelligence, EURECOM.

2019: Amazon Research Days (Berlin, Germany).

2019: Workshop on Recent developments in kernel methods, 2019, UCL (London, UK).

2019: Deep Learning Theory Kickoff Meeting 2019, MPI (Leipzig, Germany).

2018: Cambridge-Tübingen workshop 2018 (Tenerife, Spain).

Seminars.....

2021: NYU Center for Data Science. (NY, USA) Remote.

2021: Instituto Superior Técnico, Lisbon, Portugal (Remote).

2020: The Alan Turing Institute (London, UK).

2020: Department of Statistics, University of Oxford (Oxford, UK).

2019: The Alan Turing Institute (London, UK).

2018: Google Developer Group Reading and Thames Valley (Reading, UK).

Publications

- [1] M. Arbel, B. Terver, and J. Ponce. "PEIRA: Learning Predictive Encoders through Inter-View Regressor Alignment". In: *Advances in Neural Information Processing Systems (NeurIPS) 2026*. May 2026. arXiv: [2605.17671 \[cs.LG\]](#).
- [2] J. Bohne, I. Petrulionyte, M. Arbel, J. Mairal, and P. Polak. "Non-Stationary Functional Bilevel Optimization". In: *AISTATS 2026*. Jan. 2026. arXiv: [2601.15363 \[stat.ML\]](#).
- [3] F. El Khoury, H. Zenati, N. Kallus, M. Arbel, and A. Bibaut. "Semiparametric Efficient Bilevel Gradient Estimation". In: *arXiv preprint arXiv:2605.21341* (May 2026). arXiv: [2605.21341 \[stat.ML\]](#).
- [4] I. Petrulionyte, J. Mairal, and M. Arbel. "Functional Bilevel Optimization for Predictive Fairness". In: *arXiv preprint arXiv:2607.05098* (July 2026). arXiv: [2607.05098 \[cs.LG\]](#).
- [5] J. Ponce, B. Terver, M. Hebert, and M. Arbel. "Dual Perspectives on Non-Contrastive Self-Supervised Learning". In: *ICLR 2026*. June 2026. arXiv: [2507.01028 \[cs.LG\]](#).

- [6] P.-L. Ruhlmann, M. Arbel, F. Forbes, and P. L. C. Rodrigues. “Flow Matching Calibration for Simulation-Based Inference under Model Misspecification”. In: *ICML 2026*. July 2026. arXiv: [2509.23385 \[stat.ML\]](#).
- [7] E. Tournon, P. L. C. Rodrigues, J. Arbel, N. Varoquaux, and M. Arbel. “BlockFormer: Transformer-based inference from genomic contact maps”. In: *Advances in Neural Information Processing Systems (NeurIPS) 2026*. May 2026. arXiv: [2605.21617 \[cs.LG\]](#).
- [8] K. Vert, G. Meanti, S. Pesme, M. Arbel, and J. Mairal. “Beyond MMSE: Enhancing PnP Restoration with ProxiMAP”. In: *Advances in Neural Information Processing Systems (NeurIPS) 2026*. May 2026. arXiv: [2605.16396 \[cs.CV\]](#).
- [9] M. Arbel, D. Salinas, and F. Hutter. “EquiTabPFN: A Target-Permutation Equivariant Prior Fitted Network”. In: *Advances in Neural Information Processing Systems (NeurIPS) 2025*. 2025. DOI: [10.52202/085713-2097](#). arXiv: [2502.06684 \[cs.LG\]](#).
- [10] F. El Khoury, E. Pauwels, S. Vaiteer, and M. Arbel. “Learning Theory for Kernel Bilevel Optimization”. In: *Advances in Neural Information Processing Systems (NeurIPS) 2025*. 2025. arXiv: [2502.08457 \[cs.LG\]](#).
- [11] J. Marrie, R. Ménégau, M. Arbel, D. Larlus, and J. Mairal. “LUDVIG: Learning-Free Uplifting of 2D Visual Features to Gaussian Splatting Scenes”. In: *IEEE/CVF International Conference on Computer Vision (ICCV) 2025*. 2025. arXiv: [2410.14462 \[cs.CV\]](#).
- [12] G. Meanti, T. Ryckeboer, M. Arbel, and J. Mairal. “Unsupervised Imaging Inverse Problems with Diffusion Distribution Matching”. In: *ICCV 2025*. June 2025. arXiv: [2506.14605 \[cs.CV\]](#).
- [13] S. Pesme, G. Meanti, M. Arbel, and J. Mairal. “MAP Estimation with Denoisers: Convergence Rates and Guarantees”. In: *Advances in Neural Information Processing Systems (NeurIPS) 2025*. 2025. DOI: [10.52202/085713-2241](#). arXiv: [2507.15397 \[cs.LG\]](#).
- [14] E. Tournon, P. L. C. Rodrigues, J. Arbel, N. Varoquaux, and M. Arbel. “Simulation-based inference of yeast centromeres”. In: *Advances in Neural Information Processing Systems (NeurIPS) 2025, 3rd Workshop on Imageomic*. Aug. 2025. arXiv: [2509.00200 \[stat.ML\]](#).
- [15] M. Arbel and A. Zouaoui. “MLXP: A framework for conducting replicable experiments in Python”. In: *Proceedings of the 2nd ACM Conference on Reproducibility and Replicability*. 2024, pp. 134–144. arXiv: [2402.13831 \[cs.LG\]](#).
- [16] J. Marrie, M. Arbel, J. Mairal, and D. Larlus. *Data augmentation for training neural networks*. US Patent Application 18/634,466. Nov. 28, 2024.
- [17] J. Marrie, M. Arbel, J. Mairal, and D. Larlus. “On Good Practices for Task-Specific Distillation of Large Pretrained Visual Models”. In: *Transactions on Machine Learning Research Journal* (2024). arXiv: [2402.11305 \[cs.CV\]](#).
- [18] I. Petrulionyte, J. Mairal, and M. Arbel. “Functional Bilevel Optimization for Machine Learning”. In: *Advances in Neural Information Processing Systems (NeurIPS) 2024* (2024). arXiv: [2403.20233 \[stat.ML\]](#).
- [19] M. Arbel, R. Menegaux, and P. Wolinski. “Rethinking Gauss-Newton for learning over-parameterized models”. In: *Advances in Neural Information Processing Systems (NeurIPS) 2023*. 2023. arXiv: [2302.02904 \[cs.LG\]](#).
- [20] J. Marrie, M. Arbel, D. Larlus, and J. Mairal. “SLACK: Stable Learning of Augmentations with Cold-start and KL regularization”. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2023, pp. 24306–24314. arXiv: [2306.09998 \[cs.CV\]](#).
- [21] K. Pitas, M. Arbel, and J. Arbel. “Improving Deep Ensembles without Communication”. In: *Workshop on Advancing Neural Network Training (WANT@ NeurIPS 2023)*. 2023.
- [22] M. Arbel and J. Mairal. “Amortized implicit differentiation for stochastic bilevel optimization”. In: *International Conference on Learning Representations (ICLR) 2022*. 2022. arXiv: [2111.14580 \[math.OC\]](#).
- [23] M. Arbel and J. Mairal. “Non-Convex Bilevel Games with Critical Point Selection Maps”. In: *Advances in Neural Information Processing Systems (NeurIPS) 2022* (2022). arXiv: [2207.04888 \[math.OC\]](#).
- [24] Y. Eitan, N. Cavaglione, M. Arbel, and S. Cohen. “Fair synthetic data does not necessarily lead to fair models”. In: *NeurIPS 2022 Workshop on Synthetic Data for Empowering ML Research*. 2022.
- [25] P. Glaser, M. Arbel, S. Hromadka, A. Doucet, and A. Gretton. “Maximum Likelihood Learning of Unnormalized Models for Simulation-Based Inference”. In: *arXiv preprint arXiv:2210.14756* (2022). arXiv: [2210.14756 \[cs.LG\]](#).
- [26] A. Matthews, M. Arbel, D. J. Rezende, and A. Doucet. “Continual repeated annealed flow transport Monte Carlo”. In: *International Conference on Machine Learning*. PMLR. 2022, pp. 15196–15219. arXiv: [2201.13117 \[stat.ML\]](#).

- [27] T. Moskovitz, M. Arbel, J. Parker-Holder, and A. Pacchiano. "Towards an Understanding of Default Policies in Multitask Policy Optimization". In: *International Conference on Artificial Intelligence and Statistics (AISTATS) 2022*. 2022. arXiv: [2111.02994 \[cs.LG\]](#).
- [28] M. Arbel. "Methods for optimization and regularization of Generative Models". PhD thesis. UCL (University College London), 2021.
- [29] M. Arbel, A. G. Matthews, and A. Doucet. "Annealed Flow Transport Monte Carlo". In: *International Conference on Machine Learning (ICML) 2021*. 2021. arXiv: [2102.07501 \[stat.ML\]](#).
- [30] M. Arbel, L. Zhou, and A. Gretton. "Generalized Energy Based Models". In: *International Conference on Learning Representations (ICLR) 2021*. 2021. arXiv: [2003.05033 \[stat.ML\]](#).
- [31] P. Glaser, M. Arbel, and A. Gretton. "KALE Flow: A Relaxed KL Gradient Flow for Probabilities with Disjoint Support". In: *Advances in Neural Information Processing Systems (NeurIPS) 2021* (2021). arXiv: [2106.08929 \[stat.ML\]](#).
- [32] T. Moskovitz, M. Arbel, F. Huszar, and A. Gretton. "Efficient wasserstein natural gradients for reinforcement learning". In: *International Conference on Learning Representations (ICLR) 2021*. 2021. arXiv: [2010.05380 \[cs.LG\]](#).
- [33] T. Moskovitz, J. Parker-Holder, A. Pacchiano, M. Arbel, and M. I. Jordan. "Tactical Optimism and Pessimism for Deep Reinforcement Learning". In: *Advances in Neural Information Processing Systems (NeurIPS) 2021* (2021). arXiv: [2102.03765 \[cs.LG\]](#).
- [34] L. Thiry, M. Arbel, E. Belilovsky, and E. Oyallon. "The Unreasonable Effectiveness of Patches in Deep Convolutional Kernels Methods". In: *International Conference on Learning Representations (ICLR) 2021*. 2021. arXiv: [2101.07528 \[cs.CV\]](#).
- [35] M. Arbel, A. Gretton, W. Li, and G. Montúfar. "Kernelized Wasserstein Natural Gradient". In: *International Conference on Learning Representations (ICLR) 2020*. 2020. arXiv: [1910.09652 \[stat.ML\]](#).
- [36] T. Birdal, M. Arbel, U. Simsekli, and L. J. Guibas. "Synchronizing probability measures on rotations via optimal transport". In: *IEEE Conference on Computer Vision and Pattern Recognition (CVPR) 2020*. 2020, pp. 1569–1579. arXiv: [2004.00663 \[cs.CV\]](#).
- [37] S. Cohen, M. Arbel, and M. P. Deisenroth. "Estimating barycenters of measures in high dimensions". In: *arXiv preprint arXiv:2007.07105* (2020). arXiv: [2007.07105 \[stat.ML\]](#).
- [38] A. Korba, A. Salim, M. Arbel, G. Luise, and A. Gretton. "A non-asymptotic analysis for Stein variational gradient descent". In: *Advances in Neural Information Processing Systems (NeurIPS) 2020* 33 (2020). arXiv: [2006.09797 \[stat.ML\]](#).
- [39] M. Arbel, A. Korba, A. Salim, and A. Gretton. "Maximum mean discrepancy gradient flow". In: *Advances in Neural Information Processing Systems (NeurIPS) 2019* (2019). arXiv: [1906.04370 \[stat.ML\]](#).
- [40] M. Arbel and A. Gretton. "Kernel conditional exponential family". In: *International Conference on Artificial Intelligence and Statistics (AISTATS) 2018*. 2018. arXiv: [1711.05363 \[stat.ML\]](#).
- [41] M. Arbel, D. J. Sutherland, M. Bińkowski, and A. Gretton. "On gradient regularizers for MMD GANs". In: *Advances in Neural Information Processing Systems (NeurIPS) 2018* (2018). arXiv: [1805.11565 \[stat.ML\]](#).
- [42] M. Bińkowski, D. J. Sutherland, M. Arbel, and A. Gretton. "Demystifying mmd gans". In: *International Conference on Learning Representations (ICLR) 2018*. 2018. arXiv: [1801.01401 \[stat.ML\]](#).
- [43] D. J. Sutherland, H. Strathmann, M. Arbel, and A. Gretton. "Efficient and principled score estimation with Nyström kernel exponential families". In: *International Conference on Artificial Intelligence and Statistics (AISTATS) 2018*. PMLR. 2018. arXiv: [1705.08360 \[stat.ML\]](#).